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Enhanced piezoelectric effect in Janus group-III chalcogenide monolayers

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Piezoelectricity is a unique material property that converts mechanical energy into electricity or vice versa. Starting from the group-III monochalcogenide monolayers, we design a series of derivative Janus structures for piezoelectric materials, including Ga₂SSe, Ga₂STe, Ga₂SeTe, In₂SSe, In₂STe, In₂SeTe, GaInS₂, GaInSe₂, and GaInTe₂. Our first-principles calculations show that these Janus structures are thermodynamically and dynamically stable. They have a bandgap in the range of 0.89–2.03 eV, lower than those of the perfect monolayers, and Ga₂STe, Ga₂SeTe, In₂STe, and In₂SeTe monolayers are direct gap semiconductors. They possess piezoelectric coefficients up to 8.47 pm/V, over four times the maximum value obtained in perfect group-III monochalcogenide monolayers. Moreover, the broken mirror symmetry of these Janus structures induces out-of-plane dipolar polarization, yielding additional out-of-plane piezoelectric coefficients of 0.07–0.46 pm/V. The enhanced piezoelectric properties enable the development of these novel two-dimensional materials for piezoelectric sensors and nanogenerators. *Published by AIP Publishing.*
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Group-III monochalcogenides (MX, M = Ga, In; X = S, Se, Te), a new class of layered materials with honeycomb structures,^{1–3} possess intriguing physical and chemical properties and have attracted increasing attention. Recently, monolayer group-III monochalcogenides have been synthesized by mechanical exfoliation.^{1,4–8} These novel two-dimensional (2D) materials have a bandgap of 2.20–3.98 eV,^{9,10} and the gap decreases with increasing number of layers down to 0.97–3.05 eV for the bulk systems.^{5–18} Moreover, the group-III monochalcogenides exhibit high carrier mobilities, i.e., 80 and 215 cm² V^{−1} s^{−1} for bulk GaS and GaSe and 12 700 cm² V^{−1} s^{−1} for monolayer InSe, rendering them promising channel materials for field effect transistors.^{5,16} These nanosheets also hold great promise for photodetectors,^{7,14,19–21} catalysts for lithium batteries,^{22–25} and thermoelectric devices.²⁶

On the theoretical aspect, the atomic structures and mechanical and electronic properties of monolayer group-III monochalcogenides have been extensively investigated. First-principles calculations show that their bandgap can be engineered by strain and external electric field^{8,12,13} and realize a semiconductor-to-metal transition.²⁷ These monolayers are predicted to be ferromagnetic and half-metal under a broad range of hole-doping.^{28–30} They have proper band edge positions and are candidate photocatalysts for solar water splitting.⁹ These 2D materials also possess excellent thermoelectric properties with a theoretical figure of merit up to 2.43.³¹

Piezoelectric devices, which convert mechanical energy into electricity, have the advantages of large power density and ease of use for sensors and energy harvesting.^{32–37} A piezoelectric material must be non-centrosymmetric. Crystals based on the wurtzite structure are well-known piezoelectric

materials, such as α -quartz (piezoelectric coefficient = 2.3 pm/V), GaN (3.1 pm/V), AlN (5.1 pm/V), and ZnO (9.93 pm/V).^{38–41} Previous studies predict that the group-III monochalcogenide monolayers have normal piezoelectric coefficients up to 2.30 pm/V,⁴² comparable to the values of monolayer MoS₂ (3.37 pm/V) and *h*-BN (0.60 pm/V),⁴³ and thus are promising 2D piezoelectric materials.

To enhance the piezoelectric effect in group-III chalcogenide monolayers, here we design a series of Janus models, including Ga₂SSe, Ga₂STe, Ga₂SeTe, In₂SSe, In₂STe, In₂SeTe, GaInS₂, GaInSe₂, and GaInTe₂. These Janus monolayers have broken inversion and mirror symmetry. Our first-principles calculations show that they have large piezoelectric coefficients up to 8.47 pm/V, over four times the values of perfect monolayers (below 1.98 pm/V), and they exhibit additional out-of-plane piezoelectric coefficients of 0.07–0.46 pm/V. These Janus monolayers with enhanced piezoelectric properties represent a new class of 2D materials for nanogenerators, ultra-sensitive mechanical detectors, and consumer-touch sensors.

DFT calculations were performed by the Vienna *ab initio* Simulation Package (VASP) using a planewave basis set⁴³ with an energy cutoff of 450 eV (see Fig. S1 (supplementary material) for the convergence test), projector-augmented wave (PAW) pseudopotentials,^{44,45} and generalized gradient approximation for the exchange and correlation functionals parameterized by Perdew, Burke, and Ernzerhof (GGA-PBE).⁴⁶ For the structural optimization, band structure and phonon dispersion calculations, we used the primitive hexagonal unit cells of monolayer group-III chalcogenides and a 10 × 10 × 1 uniform **k** point mesh to sample the Brillouin zone. To calculate the piezoelectric coefficients, an orthorhombic supercell consisting of four group-III atoms and four chalcogen atoms was adopted, as illustrated in Fig. 1(a); the Brillouin zone was sampled by 11 × 18 × 1 Γ -centered Monkhorst-Pack **k** point grid. A vacuum region of 20 Å was added to the vertical direction to avoid interactions between

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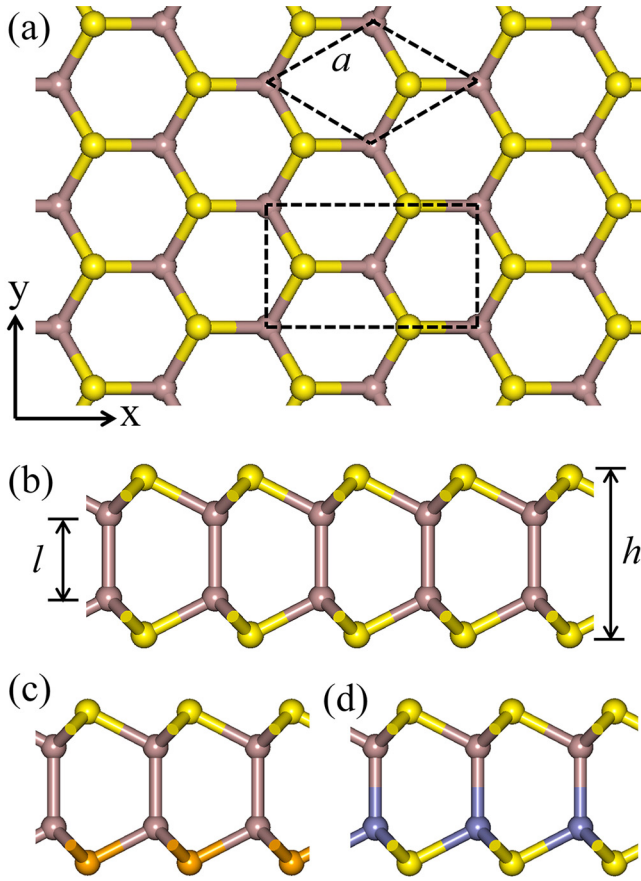


FIG. 1. Atomic structures of perfect group-III chalcogenide monolayers of (a) top view and (b) side view, and the side views of Janus structures (c) M_2XX' and (d) $MM'X_2$. The group-III atoms M and M' are shown in purple and brown, respectively. The chalcogen atoms X and X' are shown in yellow and orange, respectively.

the neighboring layers. Convergence criteria for relaxations of electronic and cell degrees of freedom were 10^{-6} eV for the total energy and 10^{-3} eV/Å for the forces on each atom. The phonon dispersions were calculated using density functional perturbation theory (DFPT)⁴⁷ as implemented in the Phonopy package.⁴⁸

As displayed in Figs. 1(a) and 1(b), monolayer group-III monochalcogenides have a layered hexagonal structure. To construct the Janus structure of monolayer GaS for instance, the bottom layer S atoms are replaced by Se atoms, yielding Ga_2SSe monolayers (see Fig. 1(c)) or the bottom Ga atoms are substituted by In atoms, forming $GaInS_2$ monolayers (see Fig. 1(d)). In such a manner, we generate a total of 9 Janus monolayers, including Ga_2SSe , Ga_2STe , Ga_2SeTe , In_2SSe , In_2STe , In_2SeTe , $GaInS_2$, $GaInSe_2$, and $GaInTe_2$. These Janus structures can be categorized into two types: M_2XX' (M = Ga, In; X, X' = S, Se, Te and $X \neq X'$) and $MM'X_2$ (M, M' = Ga, In and $M \neq M'$; X = S, Se, Te), and apparently they have broken inversion symmetry and mirror symmetry.^{49,50} For both perfect and Janus group-III monochalcogenide monolayers, the Ga–Ga, In–In, and Ga–In bond lengths are 2.47, 2.82, and 2.65 Å, respectively. The in-plane lattice constant and layer thickness of M_2XX' ($MM'X_2$) systems lie between the values of perfect MX and MX' ($M'X$) monolayers (see Table S1, [supplementary material](#)).

To characterize the thermodynamic stability of the Janus structures, we define the formation energy ΔH as

$$\Delta H = (E_{tot} - n_1 \times E_1 - n_2 \times E_2 - n_3 \times E_3)/n, \quad (1)$$

where E_{tot} is the energy of the Janus monolayer M_2XX' ($MM'X_2$); E_1 , E_2 , and E_3 are the energies of a M, X, and X' (M, M', and X) atoms in their solid phases, respectively; n_1 , n_2 , and n_3 are the numbers of atoms of each element in the supercell, and n is the total number of atoms. Our Janus models have ΔH of -0.62 to -0.33 eV (see Table I), close to the values of perfect monolayers (-0.66 to -0.34 eV), revealing that formation of the Janus group-III chalcogenide monolayers is energetically favorable. The dynamic stability of these Janus derivatives is confirmed by their phonon dispersions without imaginary bands as shown in Figs. 2(a) and S2 ([supplementary material](#)).

The electronic band structures of the perfect and Janus group-III chalcogenide monolayers are presented in Figs. 2(b) and S3 ([supplementary material](#)). Generally speaking, the Janus derivatives are still semiconductors with a bandgap ranging from 0.89 to 2.03 eV, slightly reduced with respect to those of perfect monolayers (1.32–2.36 eV, see Table I). In particular, Ga_2STe , Ga_2SeTe , In_2STe , and In_2SeTe monolayers have a direct gap with the conduction band minimum and valence band maximum located at the Γ point, while the other Janus structures as well as perfect monolayers are all indirect semiconductors.

To calculate the piezoelectric coefficients of the Janus group-III chalcogenide monolayers, we adopt the method developed by Duerloo *et al.*,⁴³ by evaluating the elastic stiffness coefficients and the change of polarization under various uniaxial strains of the Janus group-III chalcogenide monolayers. In particular, the in-plane elastic stiffness coefficients are obtained as

$$C_{11} = \frac{1}{A_0} \frac{\partial^2 U}{\partial \varepsilon_{11}^2}, \quad C_{22} = \frac{1}{A_0} \frac{\partial^2 U}{\partial \varepsilon_{22}^2}, \quad C_{12} = \frac{1}{A_0} \frac{\partial^2 U}{\partial \varepsilon_{11} \partial \varepsilon_{22}}, \quad (2)$$

where U is the energy of a Janus system; ε_{11} and ε_{22} are the strains applied to the two in-plane directions; and A_0 is the area of the unit cell at equilibrium structure. The strains ε_{11} and ε_{22} are taken from -0.006 to 0.006 with a step of 0.002 . Under each strain, the ionic positions of the Janus models are fully relaxed and the obtained C_{11} , C_{22} , and C_{12} , referred to as relaxed-ion stiffness coefficients, are experimentally relevant.^{42,43,51} For comparison, we also calculate the coefficients by fixing the atomic positions under a given strain, namely, the clamped-ion coefficients. As shown in Table I, the Janus monolayers have relaxed-ion coefficients C_{11} and C_{12} of 46–90 and 14–25 N/m, respectively ($C_{11} = C_{22}$ for the hexagonal lattice), comparable to those of perfect monolayers (55–88 and 6–21 N/m). The clamped-ion method, on the other hand, gives much larger values $C_{11} = 70$ –112 and $C_{12} = 24$ –34 N/m (see Table S1, [supplementary material](#)). It was shown for 2D materials that the relaxed-ion elastic stiffness coefficients are in better agreement with the experimental values than the clamped-ion ones.^{43,52,53} Thus, we expect that the piezoelectric coefficients calculated from the relaxed-ion method would be more reliable.

TABLE I. The calculated structural parameters (a , h , and l) defined in Fig. 1, formation energy (ΔH), bandgap (E_g), relaxed-ion elastic stiffness coefficients (C_{11} and C_{12}), Poisson ratio ($\nu_{\perp}$), and piezoelectric coefficients (e_{11} , e_{31} , d_{11} , and d_{31}) of perfect and Janus group-III chalcogenide monolayers. $\nu_{\perp}$ is equal to $\Delta h/h = -\nu_{\perp}(\epsilon_{11} + \epsilon_{22})$, where Δh is the change of layer thickness under in-plane strains.

Material	a	h	l	ΔH (eV/atom)	E_g (eV)	C_{11}	C_{12}	$\nu_{\perp}$	e_{11}	e_{31}	d_{11}	d_{31}
		(Å)				(N/m)			(10^{-10} C/m)		(pm/V)	
GaS	3.64	4.66	2.48	-0.66	2.36	88	21	0.39	1.15	—	1.72	—
GaSe	3.82	4.83	2.47	-0.60	1.81	75	20	0.34	0.98	—	1.77	—
GaTe	4.14	5.03	2.47	-0.37	1.42	65	9	0.31	1.08	—	1.93	—
InS	3.94	5.19	2.82	-0.52	1.65	80	20	0.46	0.67	—	1.12	—
InSe	4.09	5.39	2.82	-0.52	1.42	55	20	0.41	0.70	—	1.98	—
InTe	4.39	5.59	2.81	-0.34	1.32	64	6	0.28	0.69	—	1.18	—
Ga ₂ SSe	3.73	4.75	2.48	-0.62	2.03	86	23	0.48	3.25	0.08	5.23	0.07
Ga ₂ STe	3.90	4.82	2.47	-0.44	0.90	90	25	0.57	1.61	0.29	2.46	0.25
Ga ₂ SeTe	3.99	4.91	2.47	-0.46	1.22	60	17	0.45	1.00	0.16	2.32	0.21
In ₂ SSe	4.02	5.28	2.82	-0.52	1.49	55	17	0.55	3.24	0.13	8.47	0.18
In ₂ STe	4.16	5.38	2.82	-0.39	0.89	89	19	0.50	1.34	0.27	1.91	0.25
In ₂ SeTe	4.24	5.49	2.82	-0.42	1.15	69	14	0.45	2.59	0.11	4.73	0.13
GaInS ₂	3.78	4.96	2.65	-0.55	1.74	60	17	0.34	3.60	0.30	8.33	0.38
GaInSe ₂	3.96	5.14	2.65	-0.54	1.36	46	19	0.27	0.86	0.29	3.19	0.46
GaInTe ₂	4.26	5.33	2.65	-0.33	1.16	46	20	0.27	0.74	0.21	2.99	0.32

Based on the modern theory of polarization,⁵⁴ we then calculate the linear piezoelectric coefficients (d_{ijk}) of the Janus group-III chalcogenide monolayers by evaluating the change of polarization under uniaxial strains (ϵ_{ijk}) as follows:

$$e_{ijk} = \frac{\partial P_i}{\partial \epsilon_{jk}}, \quad (3)$$

$$d_{ijk} = \frac{\partial P_i}{\partial \sigma_{jk}}, \quad (4)$$

where P_i is the polarization vector; ϵ_{ijk} and σ_{ijk} are the strain and stress, respectively; i , j , and k indices correspond to the x , y , and z directions. It was shown that the number of non-zero and unique piezoelectric coefficients is restricted by the symmetry elements of a crystal. For perfect group-III chalcogenide monolayers with D_{3h} point group symmetry, the only unique piezoelectric coefficients are e_{11} and d_{11} using the Voigt notation, which characterize the in-plane piezoelectric properties of a 2D material.⁴² For the Janus derivatives with D_3 point group symmetry, the unique piezoelectric coefficients are e_{11} and d_{11} , as well as the out-of-plane coefficients e_{31} and d_{31} .^{50,55,56} These quantities are related to the elastic stiffness coefficients as

$$e_{11} = d_{11}(C_{11} - C_{12}), \quad (5)$$

$$e_{31} = d_{31}(C_{11} + C_{12}). \quad (6)$$

To obtain e_{11} , we calculate the polarization vector P_1 of the perfect and Janus monolayers under various strains ϵ_{11} in the x direction of the orthorhombic supercell (see Fig. 1(a)); ϵ_{11} is taken from -0.01 to 0.01 with a step of 0.005; e_{11} is derived as the slope of the linear relation of P_1 v.s. ϵ_{11} according to Eq. (4). For e_{31} , the out-of-plane polarization vector P_3 is computed under various strains ϵ_{11} , and e_{31} is obtained by linear fitting P_3 v.s. ϵ_{11} . For each strain value, the atomic positions of the material system are either relaxed or fixed, giving the relaxed- and clamped-ion piezoelectric coefficients, respectively.

Following the above procedures, we derive the piezoelectric coefficients of the perfect and Janus group-III chalcogenide monolayers as presented in Table I, Fig. 3(a), and Table S1 (supplementary material). The Janus structures have relaxed-ion d_{11} of 1.91–8.47 pm/V, significantly enhanced with regard to the perfect monolayers (1.12–1.98 pm/V), mainly due to their larger change of polarization under strains.⁵⁶ Specifically, the Janus systems have e_{11} of 0.74– 3.60×10^{-10} C/m, while perfect monolayers have lower

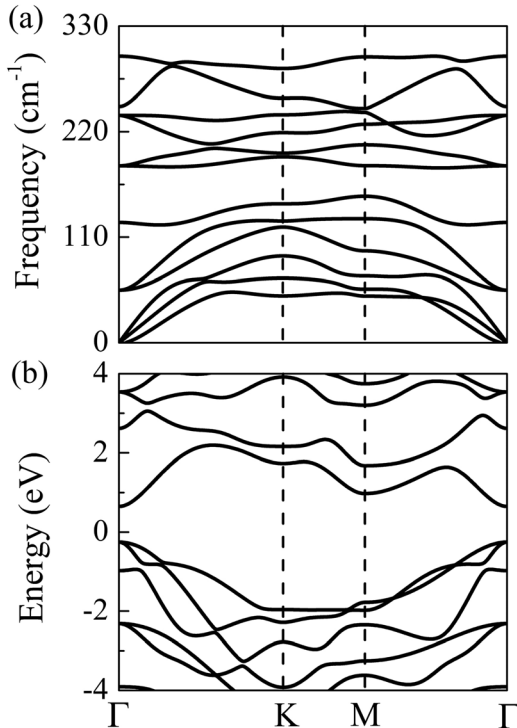


FIG. 2. (a) Phonon dispersion and (b) electronic band structure of the Janus Ga₂STe monolayer. The Fermi level of the band structure is shifted to zero.

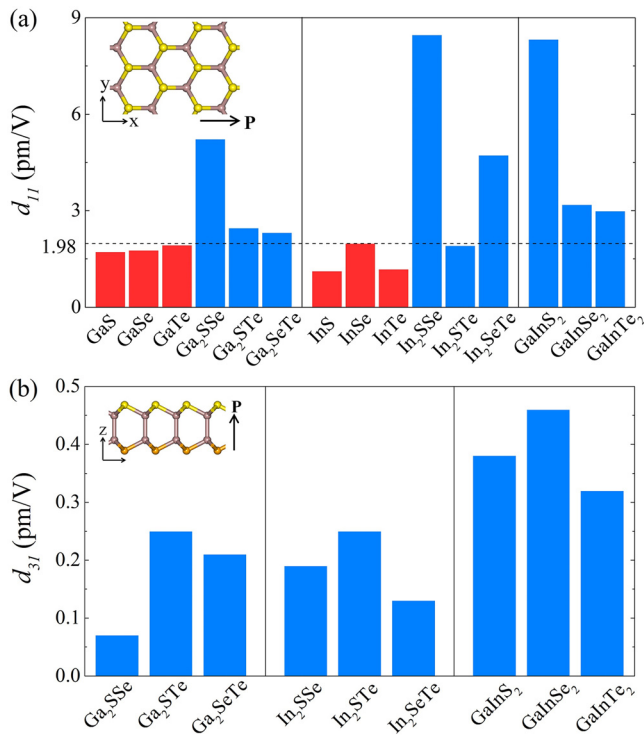


FIG. 3. Relaxed-ion piezoelectric coefficients (a) d_{11} and (b) d_{31} of Janus group-III chalcogenide monolayers. The red columns are the values of perfect monolayers, and the dashed line indicates d_{11} of InSe monolayer. The upper left insets illustrate the coordinate system and direction of polarization.

e_{11} of $0.67\text{--}1.15 \times 10^{-10}$ C/m (see Table I and Fig. S4 (supplementary material)). More noticeably, d_{11} attains 8.47 and 8.33 pm/V for In₂SSe and GaInS₂ monolayers, respectively, over four times the highest value achieved in perfect monolayers (1.98 pm/V for the InSe sheet). Compared to the recently emerging 2D piezoelectric materials such as MoS₂ and *h*-BN with $d_{11} = 3.73$ and 0.60 pm/V, respectively,³⁰ the Janus group-III chalcogenide monolayers have even larger in-plane piezoelectric coefficients by several folds, and thus, they are suitable for 2D piezoelectric sensors and nanogenerators. Note that the clamped-ion piezoelectric coefficients for the perfect and Janus group-III chalcogenide monolayers are all larger than the relaxed-ion values by 1–5 times. According to the previous studies on monolayer MoS₂, the predicted piezoelectric coefficients by the relaxed-ion method agree well with the experimental measurement, while the clamped-ion method overestimates the piezoelectric effect.^{51,57} Thus, we expect that the relaxed-ion piezoelectric coefficients of our Janus models are experimentally relevant.

Distinct from perfect group-III chalcogenide monolayers, the Janus derivatives with broken mirror symmetry have nonzero out-of-plane piezoelectric coefficients e_{31} and d_{31} , i.e., a stress or strain along the in-plane direction would induce the change of polarization perpendicular to the basal plane. The values of e_{31} and ϵ_{11} still follow the linear relation (see Fig. S5, supplementary material). The Janus monolayers have d_{31} in the range of 0.07–0.46 pm/V, as shown by Table I and Fig. 3(b). Interestingly, the MM'X₂ systems have larger d_{31} (0.32–0.46 pm/V) than those of the M₂X'X monolayers (0.07–0.25 pm/V). The highest d_{31} value (0.46 pm/V) is achieved in the GaInSe₂ monolayer, comparable to those of

some advanced piezoelectric materials such as bulk boron nitride⁵⁸ and GaN⁵⁹ in the wurtzite phase (0.33 pm/V and 0.96 pm/V, respectively). The appreciable out-of-plane piezoelectric effect would endow these Janus monolayers multiple functions for piezoelectric applications.

In summary, we design group-III chalcogenide monolayers with Janus structures for piezoelectric materials, including Ga₂SSe, Ga₂STe, Ga₂SeTe, In₂SSe, In₂STe, In₂SeTe, GaInS₂, GaInSe₂, and GaInTe₂. Our first-principles calculations show that these Janus structures are energetically and dynamically stable. Their bandgaps are slightly reduced compared to those of perfect monolayers, ranging from 0.89 to 2.03 eV, and Ga₂STe, Ga₂SeTe, In₂STe, and In₂SeTe monolayers become direct gap semiconductors. These Janus derivatives have large in-plane piezoelectric coefficients of 1.91–8.47 pm/V, significantly enhanced with regard to perfect monolayers due to the larger change of polarization under in-plane strains for the Janus systems. In addition, the broken mirror symmetry of the Janus structures gives rise to out-of-plane piezoelectric coefficients of 0.07–0.46 pm/V. As a remark, Janus structures of 2D materials have been synthesized in experiment. For instance, Janus graphene was fabricated by single-sided chlorination with the other face projected by substrate, and then peeling off the substrate and modified the other side by aryl/oxygen-functional groups.⁶⁰ Encouraged by these experimental successes, we anticipate that our proposed Janus group-III chalcogenide monolayers, with reasonable energetic and dynamic stability, could also be fabricated by similar routes in near future. The enhanced piezoelectric effect and favorable semiconducting properties of these novel 2D materials would advance the development of ultra-sensitive detectors, nanogenerators, low-power electronics, and nanoscale electromechanical systems.

See supplementary material for convergence test of the energy cutoff, complete phonon dispersions and electronic band structures, in-plane and out-of-plane polarization, and clamped-ion elastic stiffness coefficients and piezoelectric coefficients of perfect and Janus group-III chalcogenide monolayers.

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